

FLINT BISHOP, MI, USA

WMO: 726370

Lat: 42.967N

Lon: 83.749W

Elev: 235

StdP: 98.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.6	-22.4	0.5	-17.4	-19.7	0.7	-14.8	11.8	-3.0	10.8	-4.9	2.9	270	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.1	22.8	30.6	22.2	29.1	21.2	24.4	29.7	23.4	28.5	22.5	27.3	4.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	27.7	21.8	16.9	26.3	20.9	16.0	25.3	74.8	29.8	70.8	28.6	67.3	27.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	8.1	-23.0	34.6	3.6	1.6	-25.6	35.7	-27.7	36.6	-29.8	37.5	-32.4	38.6		
(5)			-23.3	26.2	3.6	1.1	-25.8	27.0	-27.9	27.7	-29.9	28.3	-32.5	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-4.8	-3.6	1.5	8.3	14.6	20.0	22.1	21.1	17.2	10.7	4.2	-1.4		
	DBStd	10.71	6.43	6.16	6.33	5.12	4.80	3.89	3.29	3.15	4.25	4.86	5.19	5.52		
	HDD10.0	1812	458	382	274	92	11	0	0	0	2	52	186	355		
	HDD18.3	3687	717	615	522	304	137	27	6	10	71	242	424	612		
	CDD10.0	1527	1	1	12	39	153	301	374	344	217	72	12	1		
	CDD18.3	359	0	0	1	2	22	77	121	97	37	4	0	0		
	CDH23.3	3376	0	0	6	28	228	738	1183	816	339	37	1	0		
	CDH26.7	1036	0	0	1	3	57	239	406	233	91	6	0	0		
(14) Wind	WSAvg	3.9	4.5	4.5	4.4	4.6	3.9	3.4	3.1	2.9	3.2	3.8	4.2	4.2		
(15) Precipitation	PrecAvg	779	42	37	52	77	78	79	77	84	81	60	61	50		
	PrecMax	1153	102	102	110	164	208	167	238	280	276	167	144	118		
	PrecMin	459	7	4	14	18	9	16	18	8	7	8	15	11		
	PrecStd	140	22	22	24	35	45	37	43	46	53	28	30	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.5	23.3	26.5	30.5	33.3	34.6	33.5	32.0	27.2	20.3	15.7		
		MCWB	11.7	10.9	16.4	16.6	20.7	22.8	24.2	23.3	22.4	19.9	14.3	13.2		
	2%	DB	9.8	10.5	18.2	23.1	28.2	31.3	32.3	31.0	29.2	23.7	16.8	11.4		
		MCWB	7.9	8.1	12.9	14.9	19.5	22.2	23.2	22.8	20.8	17.2	12.7	9.3		
	5%	DB	6.5	7.4	14.5	20.3	26.1	29.5	30.6	29.1	27.1	21.5	14.9	8.6		
		MCWB	4.6	4.8	10.6	13.1	18.6	21.3	22.4	21.6	19.9	16.4	11.4	6.6		
	10%	DB	3.5	4.4	11.1	17.4	23.5	27.6	28.9	27.7	25.1	19.0	12.4	5.9		
		MCWB	2.2	2.5	7.3	11.5	17.0	20.1	21.3	20.8	18.8	14.6	9.3	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.2	16.9	19.0	22.4	24.8	26.1	25.3	24.0	21.1	15.7	13.3		
		MCDB	12.9	13.6	22.3	24.5	28.3	30.9	32.3	30.7	29.7	25.8	18.3	15.4		
	2%	WB	8.3	8.2	13.9	16.2	20.8	23.5	24.7	24.0	22.4	19.0	13.8	9.9		
		MCDB	9.4	10.2	17.4	20.8	26.1	29.4	30.1	28.4	26.9	22.2	16.2	11.3		
	5%	WB	4.7	4.9	10.9	14.3	19.6	22.4	23.6	23.0	21.1	17.0	11.7	6.7		
		MCDB	6.3	6.9	13.9	19.1	24.4	27.8	28.7	27.3	25.1	20.5	14.3	8.4		
	10%	WB	2.4	2.6	7.6	12.0	17.9	21.3	22.5	21.9	19.9	15.2	9.4	3.9		
		MCDB	3.6	4.4	10.7	17.0	22.8	26.0	27.2	26.1	24.0	18.4	12.2	5.5		
(35) Mean Daily Temperature Range	MDBR		7.5	8.4	9.7	11.4	11.7	11.5	11.7	11.2	11.8	10.2	8.2	6.6		
	5% DB	MCDBR	9.6	10.8	13.7	15.2	13.8	13.1	12.8	12.5	13.8	13.4	11.2	9.3		
		MCWBR	8.5	8.9	9.3	9.0	7.5	6.2	5.9	5.8	6.7	8.1	8.2	7.8		
	5% WB	MCDBR	9.2	10.1	12.5	13.4	12.2	11.5	11.2	10.5	11.2	11.6	10.3	8.9		
		MCWBR	8.6	8.8	9.2	9.3	7.4	6.2	6.0	5.7	6.4	7.9	8.5	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.319	0.341	0.380	0.405	0.430	0.418	0.413	0.380	0.348	0.316	0.295		
	taud		2.360	2.286	2.360	2.276	2.260	2.219	2.267	2.275	2.356	2.436	2.478	2.414		
	Ebn at Noon		837	873	900	885	869	845	851	843	847	833	813	806		
	Edh at Noon		80	102	108	127	133	139	132	126	108	88	72	70		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.35	3.56	4.58	5.36	6.04	6.09	5.22	4.18	2.67	1.64	1.16		
	RadStd		0.14	0.30	0.37	0.40	0.47	0.36	0.29	0.21	0.39	0.25	0.21	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+51
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page